AQ10.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Let $f(x, y)$ be defined in a neighbourhood of (a, b). When is (a, b) a critical point of f?

☐ $f(a, b) = 0$.

☐ $f_x(a, b) = 0$.

☐ $f_x(a, b) = 0 = f_y(a, b)$.

☐ f_x does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x(a, b) = 0 = f_y(a, b)$.

f_x does not exist.

1 point

Consider $f(x, y) = \sqrt{5x^2 + 3y + 2}$.

☐ Any point on the line $x + y = 1$ in the domain of definition of f satisfies $f_y = 0$.

☐ There is no (x, y) such that $f_y = 0$.

☐ Solutions of $5x^2 + 3y + 2 = 0$ are critical points of f.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There is no (x, y) such that $f_y = 0$.

Solutions of $5x^2 + 3y + 2 = 0$ are critical points of f.

1 point

A critical point

☐ is a point of maximum.

☐ may be a saddle point.

☐ may be a point of local minimum.

☐ can be obtained by equating f_x and f_y to 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

may be a saddle point.

may be a point of local minimum.

can be obtained by equating f_x and f_y to 0.

1 point

Let $f(x, y) = x^2 y$. Then

☐ $(0, 0)$ is a critical point.

☐ $(0, y)$ for $y \in R$ are critical points.

☐ $f_x > 0$ at the critical points.

☐ $f_y = 0$ at the critical points.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(0, 0)$ is a critical point.

$(0, y)$ for $y \in R$ are critical points.

$f_y = 0$ at the critical points.

1 point

For $f(x, y) = x^3 + 2xy - 2x - 4y$, which of the following is/are true?

☐ $(2, -5)$ is a critical point.

☐ $(2, 5)$ is a critical point.

☐ $f_y = 0$ is a straight line parallel to the y-axis.

☐ The system of equations $f_x = 0$, $f_y = 0$ has a unique solution.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(2, -5)$ is a critical point.

$f_y = 0$ is a straight line parallel to the y-axis.

The system of equations $f_x = 0$, $f_y = 0$ has a unique solution.

1 point

Choose the correct statement(s) about the function $f(x, y) = xy^2 + 2x$.

☐ $f_x = 0$ only for 2 values of x.

☐ $f_y = 0$ along both x and y axes.

☐ There are no critical points of f.

☐ The point $(0, 0)$ is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f_y = 0$ along both x and y axes.

There are no critical points of f.

1 point

For $f(x, y) = \frac{x^3}{3} + y^2 + 2xy - 6x - 3y + 4$f(x, y) = 3x3 + y2 + 2xy − 6x − 3y + 4

☐

$f_y = 0$fy = 0 is a straight line passing through the origin.

☐

$(-1, \frac{5}{2})$(−1, 25) and $(3, -\frac{3}{2})$(3, −23) are critical points.

☐

$f_x(-1, \frac{5}{2}) = 0$fx(−1, 25) = 0.

☐

$f_y(3, -\frac{3}{2}) = 2$fy(3, −23) = 2.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(-1, \frac{5}{2})$(−1, 25) and $(3, -\frac{3}{2})$(3, −23) are critical points.

$f_x(-1, \frac{5}{2}) = 0$fx(−1, 25) = 0.

1 point

Choose the correct statement(s) about $f(x, y) = x^4 + y^4$f(x, y) = x4 + y4.

☐ There are no critical points.

☐ There is exactly one critical point.

☐

$(0, 0)$(0, 0) is a point of global minimum.

☐

$(0, 0)$(0, 0) is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:

There is exactly one critical point.

$(0, 0)$(0, 0) is a point of global minimum.

$(0, 0)$(0, 0) is a point of local minimum.

Level 2

1 point

Let $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$f(x, y) = x2 − 2xy + 4y2 − 4x − 2y + 2. Then

☐

$(3, 1)$(3, 1) is a critical point.

☐

$(2, 1)$(2, 1) is a critical point.

☐

Value of ff at $(2, 1)$(2, 1) is -4.

☐

Value of ff at $(3, 1)$(3, 1) is -9.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(3, 1)$(3, 1) is a critical point.

Value of ff at $(2, 1)$(2, 1) is -4.

1 point

Let $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$f(x, y) = x2 − 2xy + 4y2 − 4x − 2y + 2 be defined on the box $0 \le x \le 2$0 ≤ x ≤ 2 and $0 \le y \le 1$0 ≤ y ≤ 1. Then which of the following options is (are) true?

☐

Along the line joining $(0,0)$ and $(2,0)$, the minimum is attained at $(2,0)$.

☐

$(0,1)$ is a point of extremum along the line joining $(0,1)$ and $(2,1)$.

☐

$f(0,1) = 1$

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Along the line joining $(0,0)$ and $(2,0)$, the minimum is attained at $(2,0)$.

$(0,1)$ is a point of extremum along the line joining $(0,1)$ and $(2,1)$.

1 point

Let $f(x,y) = \sin x \cos y$. Then

☐

$\left((2k+1)\frac{\pi}{2}, k\pi\right)$ for $k \in \mathbb{N}$ are critical points of f.

☐

$\left((2k+1)\frac{\pi}{2}, (2k+1)\frac{\pi}{2}\right)$ for $k \in \mathbb{N}$ are critical points of f.

☐

$(k\pi, k\pi)$ for $k \in \mathbb{N}$ are critical points of f.

☐

$\left(k\pi, (2k+1)\frac{\pi}{2}\right)$ for $k \in \mathbb{N}$ are critical points of f.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\left((2k+1)\frac{\pi}{2}, k\pi\right)$ for $k \in \mathbb{N}$ are critical points of f.

$\left(k\pi, (2k+1)\frac{\pi}{2}\right)$ for $k \in \mathbb{N}$ are critical points of f.

1 point

Let $f(x,y) = e^{-x^2 + y - \frac{y^3}{3}}$. Then

☐

$f_x(1,0) = -\frac{2}{e}$.

☐

$f_x(0,0) = 1$.

☐

$f_y(0,-1) = 0$.

☐

$f_y(1,0) = e$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f_x(1,0) = -\frac{2}{e}$.

$f_y(0,-1) = 0$.

1 point

Let $f(x,y) = e^{-x^2 + y - \frac{y^3}{3}}$. Then

☐

$(0,1)$ is a critical point.

☐

$(0,-1)$ is a critical point.

☐

$(1, 1)$(1, 1) is a critical point.

☐

$(1, -1)$(1, −1) is a critical point.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(0, 1)$(0, 1) is a critical point.

$(0, -1)$(0, −1) is a critical point.

1 point

Find three positive numbers whose sum is 12 such that the sum of their squares is minimum. The mathematical expression that best defines this problem is

○

minimize $x^2 + y^2 + z^2$x2 + y2 + z2 such that $x + y + z = 12$x + y + z = 12, $x, y, z > 0$x, y, z > 0.

○

minimize $x + y + z$x + y + z such that $x^2 + y^2 + z^2 = 12$x2 + y2 + z2 = 12, $x, y, z \geq 0$x, y, z ≥ 0.

○

minimize $x^2 + y^2 + z^2$x2 + y2 + z2 such that $x + y + z = 12$x + y + z = 12, $x, y, z \geq 0$x, y, z ≥ 0.

○

minimize $x + y + z$x + y + z such that $x^2 + y^2 + z^2 = 12$x2 + y2 + z2 = 12, $x, y, z \geq 0$x, y, z ≥ 0.

No, the answer is incorrect.
Score: 0
Accepted Answers:

minimize $x^2 + y^2 + z^2$x2 + y2 + z2 such that $x + y + z = 12$x + y + z = 12, $x, y, z > 0$x, y, z > 0.

Check Answers

Your score is: 0/14